

connection between the binary expansion of a number and a convenient way to multiply with only minimal knowledge of the multiplication table. This is another reason to always keep learning: you never know when some apparently useless factoid may form the basis for a powerful algorithm.

Euclid's Algorithm

The ancient Greeks gave many gifts to humanity. One of their greatest was theoretical geometry, which was rigorously compiled by the great Euclid in his 13 books called the *Elements*. Most of Euclid's mathematical writing is in a theorem/proof style, in which a proposition is deduced logically from simpler assumptions. Some of his work is also *constructive*, meaning that it provides a method for using simple tools to draw or create a useful figure, like a square with a particular area or a tangent to a curve. Though the word had not been coined yet, Euclid's constructive methods were algorithms, and some of the ideas behind his algorithms can still be useful today.

Doing Euclid's Algorithm by Hand

Euclid's most famous algorithm is commonly known as *Euclid's algorithm*, though it is only one of many that he wrote about. Euclid's algorithm is a method for finding the greatest common divisor of two numbers. It is simple and elegant and takes only a few lines to implement in Python.

We begin with two natural (whole) numbers: let's call them a and b . Let's say that a is larger than b (if it's not, just rename a to b and rename b to a , and then a will be larger). If we divide a/b , we'll get an integer quotient and an integer remainder. Let's call the quotient q_1 , and the remainder c . We can write this as follows:

$$a = q_1 \times b + c$$

For example, if we say that $a = 105$ and $b = 33$, we find that $105/33$ is 3, remainder 6. Notice that the remainder c will always be smaller than both a and b —that's how remainders work. The next step of the process is to forget about a , and focus on b and c . Just like before, we say that b is larger